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TRANSCATHETER AORTIC VALVE REPLACEMENT APPARATUS HAVING BILATERAL LOCKING FUNCTION

Abstract

A transcatheter aortic valve replacement apparatus having a bilateral locking function includes a self-expanding stent, a valve skirt, a valve leaflet, an elastic positioning member, and an outflow end release member; a through-hole is defined in the self-expanding stent, the self-expanding stent is provided with an inflow end and an outflow end, and the sidewall of the self-expanding stent is a grid structure; the valve skirt is connected to the sidewall of the through-hole; the valve leaflet is connected to the inner side of the valve skirt, and the sidewall of the self-expanding stent adjacent to the outflow end is defined by a plurality of outflow end grids; the elastic positioning member is provided with a U-shaped portion and connecting arms located at two ends of the U-shaped portion, and the connecting arms are connected to the self-expanding stent or to the outflow end release member, forming first connection points.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a National Stage application of PCT/CN2023/091878, filed on Apr. 28, 2023, which claims priority to Chinese patent application No. 202210473085X, filed on Apr. 29, 2022, the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a technical field of structural design of an apparatus for interventional treatment with a prosthetic heart valve, in particular, to a transcatheter aortic valve replacement apparatus having bilateral locking function.

BACKGROUND

[0003] With the aging of the population, the incidence rate of valvular heart disease is obviously increased. At present, conventional surgical treatment is still the treatment of choice for most patients with severe valvular lesions, but there is a high risk of high trauma, postoperative mortality rate and complications. In recent years, transcatheter valve implantation/repair has gradually matured and been widely used, especially transcatheter aortic valve implantation (TAVR/TAVI), which has sufficient evidence-based basis and greatly reduces trauma, and has been recommended by European and American guidelines for the management of valvular heart disease. This is a milestone in interventional treatment of valvular heart diseases.

[0004] Transcatheter aortic valve implantation (TAVI) is a new technique for interventional placement of prosthetic aortic valves. It was first reported by Dr. Cribier in France in 2002, offering hope for the treatment of patients with severe aortic stenosis (AS) who lost surgical opportunity (e.g., patients over 80 years old) and opening a new page in the history of cardiovascular intervention. Over the next decade, with the improvement of instruments and the accumulation of experience, the TAVI technique are being refined. More than 500 cardiac centers in nearly forty countries have practiced this technique, with a total of over 150000 cases. In particular, the efficacy, feasibility, and safety of TAVI have been demonstrated after a series of registration studies and randomized controlled trials. TAVI technique has become the treatment of choice for patients with severe AS who cannot undergo surgical valve replacement. Clinically, the prosthetic bioprosthetic valves used for TAVI primarily includes an Edwards Sapien (available from Edwards Lifesciences, USA) which is to be implanted by balloon expansion and a Core Valve (available from Medtronic, USA) which is to be implanted by self-expansion. TAVI technique has made impressive progress internationally. TAVI technique has been preliminarily applied in China and has a promising prospect. At present, the domestic heart valve instrument market is a market highly monopolized by foreign brands. Foreign-funded enterprises such as Edwards Lifesciences, Medtronic, LivaNova (acquired by Solin), St. Jude Medical (acquired by Abbott) and On-X account for about 85% of the market share. Among them, Edwards Lifesciences and Medtronic have complete product lines from mechanical valves and bioprosthetic valves to transcatheter

interventional valves. At the same time, a number of domestic enterprises of medical instrument have emerged in China. At present, three domestic transcatheter aortic valves have been approved by China's CFDA, and are a Venus-A available from Venus Medtech, a J-valve available from Suzhou Jiecheng's and a VitaFlow available from MicroPort CardioFlow, respectively. However, none of the domestic enterprises has an absolute leading advantage.

[0005] According to the statistical analysis of the echocardiography database of hospital patients, in patients aged 65 to 74 years (49995 cases) and patients over 75 years (34671 cases), the detection rates of moderate or severe aortic regurgitation (AR) were 2.12% and 2.85% respectively, the detection rates of moderate or severe aortic stenosis (AS) were 0.75% and 0.89% respectively, and the detection rates of severe aortic regurgitation (SAR) and severe aortic stenosis (SAS) were 0.52% VS 0.95% and 0.54% VS 0.57% respectively, in both aged groups. It can be seen that Chinese elderly people tend to have aortic regurgitation in degenerative aortic valvular diseases. There are some differences between the patients with aortic valve disease in China and those in western countries: (1) The proportion of patients with bicuspid aortic valve in China is high. The proportion of patients with bicuspid aortic valve is in a range of 40%~50%, which is much higher than the 1.6%~9.3% in western countries. Multiple large-scale TAVR clinical studies in Western countries have listed the bicuspid aortic valve as an exclusion criterion. (2) The calcification degree of aortic valve is higher in China. (3) In China, there are more patients with aortic valve regurgitation than with aortic valve stenosis. (4) The inner diameter of femoral artery is smaller, in China, the average inner diameter of femoral artery in the candidate cases of TAVR was 6.5 mm.

[0006] The Venus-A available from Venus Medtech and the VitaFlow available from MicroPort CardioFlow adopt percutaneous route and self-expansion stent design, and are mainly used for treating aortic valve stenosis. Although the J-valve available from Suzhou Jiecheng also adopts a design of a self-expansion stent and has three anchoring devices that secure the stent to the bottom of the native valve leaflet to address aortic regurgitation, the path adopted is transapical and the transapical path is more traumatic than the percutaneous path. Meanwhile, products of the Venus-A available from Venus Medtech and the J-valve available from Suzhou Jiecheng adopt porcine pericardium, and the VitaFlow available from MicroPort CardioFlow adopts bovine pericardium which has better durability than porcine pericardium. To sum up, the three aortic valve products currently on the market in China all have certain limitations in the treatment of disease or their own performance deficiencies.

[0007] In the Chinese patent application No. 2021107458307, there is provided a transcatheter aortic valve device for automatically capturing valve leaflets. The device has a stent, a valve skirt, and a positioning member. The positioning member is an extended part of the stent, and the connection between the positioning member and the stent also forms a release structure of the stent. In actual use, the connection between the stent and the positioning member is used to release the stent. In the clinical operation process, it is found that, when the above technique is used, the connection between the stent and the positioning member may affect the positioning of the positioning member in the stent release process, resulting in inaccurate positioning of the device, adverse surgical consequences, and even endangering the life safety of the patient in serious cases. In addition, as blood and cardiac tissue exert a large force on the stent, relative displacement occurs between the stent and the positioning member, thereby causing the positioning member to be pulled and affecting its lifespan.

SUMMARY

[0008] Based on the above description, the present disclosure provides a transcatheter aortic valve replacement apparatus having bilateral locking function, so as to solve the technical problems in the prior art that the positioning of the positioning member may be affected in the stent release process due to unreasonable structural design of the aortic valve product and thus the positioning of the apparatus is inaccurate.

[0009] The technical solution for solving the above technical problems is as follows.

[0010] A transcatheter aortic valve replacement apparatus having bilateral locking function, includes a self-expansion stent, a valve skirt, a valve leaflet, at least two elastic positioning members, an outflow end release member, and an inflow end release member.

[0011] A through-hole for liquid circulation is defined within the self-expansion stent. The self-expansion stent has an inflow end and an outflow end opposite to the inflow end corresponding to the through-hole. A sidewall of the self-expansion stent has a grid-shaped structure. The valve skirt is connected to a sidewall of the through-hole. The valve leaflet is connected to an inner side of the valve skirt. A sidewall of the self-expansion stent adjacent to the outflow end is formed by a plurality of outflow end grids arranged in an annular shape.

[0012] Each elastic positioning member has a U-shaped portion and connecting arms located at two ends of the U-shaped portion. Each connecting arm is connected to the outflow end. The U-shaped portion is located on an outer side of the self-expansion stent. A middle part of the U-shaped portion extends toward the inflow end and is elastically foldable.

[0013] The outflow end release member is connected to the outflow end of the self-expansion stent and located between two of the elastic positioning members. The inflow end release member is connected to the inflow end of the self-expansion stent.

[0014] Compared with the prior art, the technical solution of the present application has the following beneficial technical effects.

[0015] In the transcatheter aortic valve replacement apparatus provided in this disclosure, the elastic positioning member and the outflow end release member are provided on the stent, and the positioning of the stent and the release of the outflow end are separate in function, and thus are independent from each other and do not affect each other, which effectively avoids the effective mounting of the positioning member and even the entire self-expansion stent from being affected when the stent is released by the outflow end release member. In addition, the outflow end release member is mounted between the elastic positioning members, which effectively ensures the compactness and stability of the whole apparatus. The outflow end release member and the inflow end release member can be fixed on the delivery system at both sides, and can be more conveniently mounted inside the delivery system.

[0016] On the basis of the above technical solution, this disclosure can also be improved as follows.

[0017] Further, the connecting arm is connected to the outflow end grid or the outflow end release member at a first connection point. The outflow end release member is connected to the outflow end grid at a second connection point. At least one of the first connection point and the second connection point is staggered from a vertex of the outflow end grid.

[0018] After the above technical solution is adopted, one of the first connection point and the second connection point is separated from the vertex of the outflow end grid, and the force forcing relative displacement on the connection between the elastic positioning member and the self-expansion stent can be effectively dispersed, which can effectively protect the elastic positioning member from fracturing.

[0019] Further, an included angle between the connecting arm and a plane tangent to the corresponding outflow end grid is in a range of $0^{\circ}\sim 45^{\circ}$.

[0020] Further, the connecting arm is located on the plane tangent to the corresponding outflow end grid.

[0021] After the above technical solution is adopted, the radial pressure received by the elastic positioning member can be effectively transferred to the self-expansion stent, and then the self-expansion stent would bear such force, so as to counteract the radial force borne by the positioning member.

[0022] Further, three elastic positioning members are provided. The three elastic positioning members are evenly distributed in a circumferential direction of the self-expansion stent.

[0023] Further, repair parts are formed near two ends of the U-shaped portion. The transcatheter

aortic valve replacement apparatus further includes a repair member. Each repair part is connectable to the self-expansion stent through the repair member.

[0024] Further, the repair part and the first connection point are located at two ends of the connecting arm. The repair member has a soft sheet-like structure. The repair member is capable of wrapping or winding around the connecting arm in a length direction of the connecting arm. Two ends of the repair member corresponding to the connecting arm are fixed to the corresponding first connection point and the corresponding repair part, respectively.

[0025] After the above technical solution is adopted, the repair member is wound around the connecting arm and fixed, which further enhance the connection between the connecting arm and the self-expansion stent and the U-shaped portion. When any end of the elastic positioning member disengages from the self-expansion stent, the elastic positioning member can be restricted by the repair member to avoid further damage to the patient's body.

[0026] Further, the repair member has an elongated strip structure and one repair member is provided. An end of the repair member is capable of extending through all of the repair parts in sequence and being connected to the self-expansion stent.

[0027] Further, the repair member has an elongated strip structure and the number of the repair members is equal to the number of the elastic positioning members. An end of each of the repair members extends through the repair parts on the same U-shaped portion in sequence and is connected to the self-expansion stent.

[0028] After the above technical solution is adopted, the connection relationship between the elastic positioning member and the self-expansion stent can be repaired in time by the repair member when the elastic positioning member falls off or is connected unstably, which can prevent the patient from being further damaged.

[0029] Further, the transcatheter aortic valve replacement apparatus further includes a protective member connected to an outer side of a middle part of the U-shaped portion by suturing, wrapping, or winding.

[0030] Further, the repair member and the protective member are polyester cloth or a bovine pericardium.

[0031] After the above technical solution is adopted, the U-shaped portion is covered by the protective member during use, which can effectively prevent the U-shaped portion from scratching the cardiac tissue or blood vessel wall.

[0032] Further, the outflow end release member includes a release portion and support arms located on two sides of the release portion. The support arm is connected to the self-expansion stent at the second connection point. Two second connection points of the same outflow end release member are located at vertexes of different outflow end grids. The release portion extends away from the outflow end in the first direction.

[0033] After the above technical solution is adopted, the outflow end release member and the elastic positioning member respectively extend in two directions, so that the interplay between the two is further reduced.

[0034] Further, a release grid is enclosed by the support arms of each outflow end release member and adjacent outer sides of two outflow end grids. An end of the valve leaflet adjacent to the inflow end is sutured to the valve skirt in a circumferential direction, and another end of the valve leaflet is connected to the valve skirt at a position inside the release grid.

[0035] After the above technical solution is adopted, the release grid is enclosed by the support arms and the adjacent outer sides of the two outflow end grids, and the end of the valve leaflet adjacent to the outflow end is connected to the valve skirt at the position inside the release grid, so that the size of the release grid is much greater than the area available at the connection in the prior art, such that the valve leaflet is bonded to the valve skirt more conveniently and firmly, and the service life of the apparatus is prolonged.

[0036] Further, a hole size of the through-hole gradually becomes smaller and then remains

unchanged or gradually becomes larger in a first direction, and the release grid is bent toward an inner side of the through-hole in the first direction.

[0037] Further, the elastic positioning member and the self-expansion stent are formed by cutting a nickel-titanium alloy pipe in a single pass, followed by thermoforming.

[0038] After the above technical solution is adopted, the stability of the structural connection between the self-expansion stent and the elastic positioning member is ensured.

[0039] Further, a sidewall of the self-expansion stent adjacent to the inflow end is formed by a plurality of inflow end grids arranged in an annular shape. The inflow end release member includes an elliptical structure connected to a vertex of one of the inflow end grids.

[0040] After the above technical solution is adopted, since the elliptical structure has no sharp end, the outer side of the elliptical structure is a smooth curved surface, which reduces the space for the movement of the inflow end release member and prevents the vascular wall from being scratched.

[0041] Further, a penetration hole is formed at a middle part of the elliptical structure in a radial direction of the through-hole.

[0042] After the above technical solution is adopted, the whole elliptical structure is ring-shaped, compared with a solid elliptical structure, this elliptical structure is softer as a whole, and it is not easy to puncture human organs and tissues.

[0043] Further, an extension line of a short axis of the elliptical structure is parallel to an axis of the through-hole, and an extension line of a long axis of the elliptical structure is located on a plane tangent to the corresponding inflow end grid.

[0044] After the above technical solution is adopted, the inflow end grid is designed to have a vertical contraction form at the apex, which can effectively prevent the aortic vessel wall from being scratched during the mounting of the stent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0045] FIG. 1 is a perspective schematic view illustrating a structure of a transcatheter aortic valve replacement apparatus having bilateral locking function according to an embodiment of the present disclosure.

[0046] FIG. 2 is a schematic view illustrating the structure in FIG. 1 with the valve skirt and the valve leaflet omitted.

[0047] FIG. 3 is a schematic view illustrating a connection structure, with a first connection point, a second connection point coinciding with a vertex of an outflow end grid.

[0048] FIG. 4 is a schematic view illustrating a structure of an embodiment adopting a first staggered setting of the present disclosure.

[0049] FIG. 5 is a schematic view illustrating a structure of an embodiment adopting a second staggered setting of the present disclosure.

[0050] FIG. 6 is a top schematic view illustrating a structure in which a support arm and a self-expansion stent are disposed at a certain angle.

[0051] FIG. 7 is a top schematic view illustrating a structure in which a support arm and a self-expansion stent are tangent to each other.

[0052] FIG. 8 is a schematic view illustrating a structure of another embodiment having a repair part compared with the embodiment shown in FIG. 5.

[0053] FIG. 9 is a schematic view illustrating a structure of a repair part in another embodiment of the present disclosure.

[0054] FIG. 10 is a schematic view illustrating a compressed structure according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0055] In order to facilitate the understanding of the present application, a more comprehensive description of the present application will be provided below with reference to the relevant drawings. Embodiments of the present application are illustrated in the drawings. However, the present application may be implemented in many different forms and is not limited to the embodiments described herein. Rather, these embodiments are provided for the purpose of making the disclosure of the present application more thorough and comprehensive.

[0056] Unless otherwise defined, all technical and scientific terms used herein have the same meanings as those commonly understood by those skilled in the art to which the present application belongs. The terms used in the description of the present application herein are only for the purpose of describing specific embodiments and are not intended to limit the present application.

[0057] As shown in FIG. 1, the present application provides a transcatheter aortic valve replacement apparatus having bilateral locking function, including a self-expansion stent **10**, a valve skirt **20**, a valve leaflet **30**, at least two elastic positioning members **40**, and at least two outflow end release members **50**.

[0058] In the present embodiment, the self-expansion stent **10** defines a through-hole therewithin for liquid (e.g., blood) circulation. The self-expansion stent **10** has an inflow end **1a** and an outflow end **1b** opposite to the inflow end **1a**, corresponding to the through-hole. The sidewall of the self-expansion stent **10** has a grid-shaped structure. The valve skirt **20** is connected to the sidewall of the through-hole. The valve leaflet **30** is connected to the inner side of the valve skirt **20**. As shown in FIG. 2, the valve leaflet **30** is configured to allow blood to flow through the through-hole in a first direction A from the inflow end **1a** to the outflow end **1b**, and prevents blood from flowing through the through-hole in a second direction B opposite to the first direction A. The sidewall of the self-expansion stent **10** adjacent to the inflow end **1a** is formed by a plurality of inflow end grids **11** arranged in an annular shape. The sidewall of the self-expansion stent **10** adjacent to the outflow end **1b** is formed by a plurality of outflow end grids **12** arranged in an annular shape.

[0059] As shown in FIG. 2, the elastic positioning member **40** has a U-shaped portion **41** and connecting arms **42** located at two ends of the U-shaped portion **41**. The U-shaped portion **41** is located on the outer side of the self-expansion stent **10**. The middle part of the U-shaped portion **41** extends toward the inflow end **1a** and can be elastically folded. It is well known that when the self-expansion stent **10** is fed into the human body, the self-expansion stent **10** needs to be compressed. As shown in FIG. 10, the U-shaped portion **41** can be elastically folded to extend toward the outflow end.

[0060] The outflow end release member **50** is connected to the outflow end **1b** of the self-expansion stent **10** and located between the two elastic positioning members **40**. The connecting arm **42** is connected to the outflow end grid **12** or the outflow end release member **50** at a first connection point L1. The outflow end release member **50** is connected to the outflow end grid **12** at a second connection point L2.

[0061] In this embodiment, the elastic positioning of the self-expansion stent **10** and the release of the outflow end **12** are separate in function, and thus are independent from each other and do not affect each other. Therefore, it can effectively avoid that the heart valve cannot be accurately positioned due to the fact that the self-expansion stent **10** may affect the positioning and mounting of the elastic positioning member **40** during the release of the outflow end. In addition, the outflow end release member **50** is mounted between the elastic positioning members **40**, which effectively ensures the compactness and stability of the whole apparatus.

[0062] In the present application, the outflow end release member **50** includes a release portion **51** and support arms **52** located on two sides of the release portion. The second connection point L2 is formed by connecting the support arm **52** to the self-expansion stent **10**. The two second connection points L2 of the same outflow end release member **50** are located at the vertexes of different outflow end grids **12**, respectively. The release portion **51** extends away from the outflow end **1b** in the first direction.

[0063] The outflow end release member **50** may be considered as a part extending from or connected to the self-expansion stent **10**. The elastic positioning member **40** may be connected to the outflow end grid **12** of the self-expansion stent **10**, or may be connected to the two support arms **52**.

[0064] The release portion **51** is configured to be connected to a delivery system that delivers the valve apparatus into the human body. When the release portion **51** is disconnected from the delivery system, driven by the self-expansion stent **10**, the outflow end release member **50** expands outwardly, until the outflow end **1b** is released. After the mounting is completed, the outflow end release member **50** and the elastic positioning member **40** respectively extend in two directions, so that the interplay between the two is further reduced.

[0065] More preferably, a release grid **53** is enclosed by the support arms **52** of each outflow end release member **50** and the adjacent outer sides of two outflow end grids **12**. An end of the valve leaflet **30** adjacent to the inflow end **1a** is sutured to the valve skirt **20** in a circumferential direction, and another end of the valve leaflet **30** is connected to the valve skirt **20** at a position inside the release grid **53**. The release grid **53** is enclosed by the support arms **52** and the adjacent outer sides of the two outflow end grids **12**, and the end of the valve leaflet **30** adjacent to the outflow end **12** is connected to the valve skirt **20** at the position inside the release grid **53**, so that the size of the release grid **53** is much greater than the area available at the connection in the prior art, such that the valve leaflet **30** is bonded to the valve skirt **20** more conveniently and firmly, and the service life of the heart valve apparatus is prolonged.

[0066] For the connection mode between the elastic positioning member **40** and the outflow end release member **50**, the most obvious and most easily conceived mode is shown in FIG. 2 and FIG. 3, that is, the first connection point **L1** and the second connection point **L2** are both at the vertex of the outflow end grid **12**.

[0067] After the force analysis and research on the transcatheter aortic valve replacement apparatus during use, it is found that the self-expansion stent **10** and the outer sidewall of the elastic positioning member **40** are in contact with the cardiac tissue during use. Firstly, the elastic positioning member **40** needs to bear the pressure from the cardiac tissue in the radial direction of the self-expansion stent **10**. At the same time, due to the frictional force generated when the blood flows through the interior of the through-hole, and the swirling force due to uneven pressure of the blood, the self-expansion stent **10** and the elastic positioning member **40** tend to rotate relative to each other, thereby forming a rotational pulling force. In addition, because the end of the valve leaflet **30** adjacent to the outflow end **12** is connected to the valve skirt **20** at the position inside the release grid **53**, when the blood flows through the through-hole, the valve leaflet **30** is periodically opened and closed, so that the regions near the outflow end grid **12** are periodically swung, thereby forming a periodic swinging pulling force on the elastic positioning member **40**. When the first connection point **L1**, the second connection point **L2** and the vertex of the outflow end grid **12** are clustered together, the above-mentioned several forces will act on the elastic positioning member **40**, forcing the elastic positioning member **40** and the self-expansion stent **10** to move relative to each other and generating a large pulling force therebetween, which increases the risk of fracture of the elastic positioning member **40**. If both the first connection point **L1** and the second connection point **L2** coincide with the vertex of the outflow end grid **12**, the fracture risk will be greatly increased. Therefore, in the present application, the more dispersed the first connection point **L1**, the second connection point **L2** and the vertex of the outflow end grid, the less pulling force the elastic positioning member **40** would be subjected to, which effectively reduces the risk of fracture of the two ends of the elastic positioning member **40**. Preferably, at least one of the first connection point **L1** and the second connection point **L2** is staggered from the vertex of the corresponding outflow end grid **12**.

[0068] Obviously, the above staggered setting includes three cases. The first case is shown in FIG. 4, that is, the second connection point **L2** coincides with the vertex of the outflow end grid **12**, and

the first connection point L1 is staggered from such coinciding position. In this case, the connecting arm 42 can be directly connected to the support arm 52. The second case is shown in FIG. 5, that is, the first connection point L1 coincides with the vertex of the outflow end grid 12, and the second connection point L2 is staggered from such coinciding position. The third case is that both the first connection point L1 and the second connection point L2 are staggered from the vertex of the outflow end grid 12.

[0069] For each one of the connection cases as described above, one of the first connection point and the second connection point is effectively separate, so as to disperse the force and prevent the elastic positioning member 40 from fracturing.

[0070] In a preferred embodiment of the present application, three elastic positioning member 40 and three outflow end release member 50 are provided. The three elastic positioning members and the three outflow end release members 50 are evenly distributed in the circumferential direction of the self-expansion stent 10.

[0071] The relative included angle between the connecting arm 42 and the outflow end grid 12 can meet the use requirements within a certain range of the included angle. Specifically, the included angle between the connecting arm 42 and the plane tangent to the corresponding outflow end grid 12 is in a range of 0°-45°. In the top view shown in FIG. 6, the included angle between the connecting arm 42 and the plane tangent to the self-expansion stent 10 is 45°.

[0072] According to the fracture analysis of the elastic positioning member 40, it is found that the main fracture effect comes from the radial pressure. Therefore, it is preferred that the connecting arm 42 is tangent to the self-expansion stent 10. When the cardiac tissue exerts the radial pressure on the elastic positioning member 40, such structure will release the radial pressure and transfer the radial pressure to the self-expansion stent 10. As such, the self-expansion stent would bear such acting force, so as to achieve the effect that the elastic positioning member 40 can counteract the radial pressure. Therefore, in this embodiment, as shown in FIG. 7, the connecting arm 42 is located on the plane tangent to the corresponding self-expansion stent 10, that is, the included angle between the connecting arm 42 and the plane tangent to the corresponding outflow end grid 12 is 0.

[0073] Based on the above description, the present application can effectively reduce the risk of the elastic positioning member being broken by dispersing the force on the one hand, and by counteracting the force on the other hand, which greatly prolongs the service life of the heart valve.

[0074] However, in the practical use of the heart valve, with the lengthening of the use time and the aging of the material, the elastic positioning member 40 and the self-expansion stent 50 may still fracture and fall off. If the elastic positioning member 40 is allowed to move freely in the body, the elastic positioning member 40 may puncture the heart, which may cause serious irreversible damage to the patient's body. Therefore, an effective repair means needs to be provided. As shown in FIG. 8 and FIG. 9, in another preferred embodiment of the present application, repair parts 43 are formed near both ends of the U-shaped portion 41. The transcatheter aortic valve replacement apparatus further includes a repair member (not shown in the figures). The repair part 43 may be connected to the self-expansion stent 10 through the repair member. In some embodiments, the repair part 43 is a thick part having holes and formed at the ends of the U-shaped portion 41. In other embodiments, the repair parts are lugs formed at the ends of the U-shaped portion 41, which may be in a shape of a closed loop or may be a hook bent inward.

[0075] The repair part 43 and the first connection point L1 are located at two ends of the connecting arm 42. In some embodiments, the repair member has a soft sheet-like structure. The repair member can wrap or wind around the connecting arm 42 in the length direction of the connecting arm 42. The two ends of the repair member corresponding to the connecting arm are fixed to the corresponding first connection point L1 and the corresponding repair part 43, respectively.

[0076] Preferably, in the above embodiment, the repair member is a polyester cloth or a bovine pericardium.

[0077] In some other embodiments, the repair member may have an elongated strip structure, specifically, may be a rope or a nickel-titanium wire, and one repair member is provided. An end of the repair member may extend through all repair parts **43** in sequence and be connected to the self-expansion stent **10**. In some other embodiments, the number of repair members is equal to the number of elastic positioning members **40**. An end of each repair member extends through the repair parts **43** on the same U-shaped portion **41** in sequence, and is connected to the self-expansion stent **10**. The above connection may optionally be hooked, wound, or bound to the self-expansion stent **10**, so as to ensure that the U-shaped portion **41** does not disengage from the self-expansion stent **10**.

[0078] After the above technical solution is adopted, the connection relationship between the elastic positioning member **40** and the self-expansion stent **10** can be repaired in time by the repair member when the elastic positioning member **40** falls off or is connected unstably, so as to prevent the patient from being injured.

[0079] In a preferred embodiment of the present application, the transcatheter aortic valve replacement apparatus further includes a protective member **44**. The protective member **44** is connected to the outer side of the middle part of the U-shaped portion **41** by suturing, wrapping, or winding. The protective member **44** is preferably a polyester cloth or a bovine pericardium.

[0080] In the embodiment of the present application, the hole size of the through-hole gradually becomes smaller and then remains unchanged or gradually becomes larger in the first direction A. Such trumpet-shaped opening can make the self-expansion stent **10** be well fixed at the aortic valve of the human body. The release grid **53** is bent toward the inner side of the through-hole in the first direction A. As an optional connection mode, the release portion **51** has an annular structure. The delivery system and the release portion **51** are connected to each other by a strip-shaped connecting piece having a coil. The coil can pass into the other side of the release portion **51** in the second direction. The strip-shaped piece, which can be axially movable, extends through the coil to limit the position of the release portion **51**. When the strip-shaped piece is withdrawn from the coil, the release portion **51** can be disengaged from the outside of the coil under the effect of elasticity to complete the release of the outflow end. Since the release grid **53** is bent inwardly in the first direction A, the whole strip-shaped piece may be located inside the tubular body formed after the self-expansion stent is compressed, and the placement of the strip-shaped piece may prevent the cardiac tissue from being punctured.

[0081] In order to ensure the stability of the structural connection between the self-expansion stent **10** and the elastic positioning member **40**, the elastic positioning member **40** and the self-expansion stent **10** are formed by cutting a nickel-titanium alloy pipe in a single pass and then thermoforming it.

[0082] To achieve accurate release and mounting of the inflow end, as shown in FIG. 1 and FIG. 2, the transcatheter aortic valve replacement apparatus further includes a plurality of inflow end release members **60**. The inflow end release member **60** includes an elliptical structure **61** connected to the vertex of the inflow end grid **11**. Since the elliptical structure **61** has no sharp end, the outer side of the elliptical structure **61** is a smooth curved surface, which reduces the space for the movement of the inflow end release member **60** and prevents the vascular wall from being scratched. In order to further strengthen the soft characteristics of the elliptical structure, a penetration hole is formed at its middle part in the radial direction of the through-hole. In this way, the elliptical structure **61** is ring-shaped as a whole. The solid elliptical structure is very hard and protrudes outwardly relative to the inflow end, and easily punctures the cardiac tissue. The elliptical structure **61** is provided with a ring-shaped structure and cooperate with the nickel-titanium alloy material, to be more soft, so that the human organs and tissues are not easy to be punctured. In addition, both the inflow end release member **60** and the outflow end release member **50** that are ring-shaped can be fixed on the delivery system at both sides, without affecting the mounting and releasing function of the stent.

[0083] In a preferred embodiment, the extension line of the short axis of the elliptical structure **61** is parallel to the axis of the through-hole, and the extension line of the long axis of the elliptical structure **61** is located on the plane tangent to the corresponding inflow end grid **11**. In this way, the inflow end grid **11** is designed to have a vertical contraction form at the vertex, which can effectively prevent the aortic vessel wall from being scratched during the mounting of the transcatheter aortic valve replacement apparatus.

[0084] The above description are only preferred embodiments of the present disclosure and is not intended to limit the present disclosure. Any modification, equivalent replacement, improvement, etc. made within the spirit and principle of the present disclosure shall be included in the scope of protection of the present disclosure.

Claims

1. A transcatheter aortic valve replacement apparatus having bilateral locking function, comprising a self-expansion stent, a valve skirt, a valve leaflet, at least two elastic positioning members, an outflow end release member, and an inflow end release member; wherein a through-hole for liquid circulation is defined within the self-expansion stent; the self-expansion stent has an inflow end and an outflow end opposite to the inflow end, corresponding to the through-hole; and a sidewall of the self-expansion stent has a grid-shaped structure; wherein the valve skirt is connected to a sidewall of the through-hole; wherein the valve leaflet is connected to an inner side of the valve skirt; a sidewall of the self-expansion stent adjacent to the outflow end is formed by a plurality of outflow end grids arranged in an annular shape; each elastic positioning member has a U-shaped portion and connecting arms located at two ends of the U-shaped portion; wherein each connecting arm is connected to the outflow end; the U-shaped portion is located on an outer side of the self-expansion stent; and a middle part of the U-shaped portion extends toward the inflow end and is elastically foldable; and wherein the outflow end release member is connected to the outflow end of the self-expansion stent and located between two of the elastic positioning members; and the inflow end release member is connected to the inflow end of the self-expansion stent.
2. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 1, wherein the connecting arm is connected to the outflow end grid or the outflow end release member at a first connection point; the outflow end release member is connected to the outflow end grid at a second connection point; and at least one of the first connection point and the second connection point is staggered from a vertex of the outflow end grid.
3. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 1, wherein an included angle between the connecting arm and a plane tangent to the corresponding outflow end grid is in a range of $0^{\circ}\sim 45^{\circ}$.
4. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 3, wherein the connecting arm is located on the plane tangent to the corresponding outflow end grid.
5. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 1, wherein three elastic positioning members are provided, and the three elastic positioning members are evenly distributed in a circumferential direction of the self-expansion stent.
6. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 1, wherein repair parts are formed near two ends of the U-shaped portion; the transcatheter aortic valve replacement apparatus further comprises a repair member, and wherein each repair part is connectable to the self-expansion stent through the repair member.
7. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 6, wherein the repair part and the first connection point are located at two ends of the connecting arm; the repair member has a soft sheet-like structure; the repair member is capable of wrapping or winding around the connecting arm in a length direction of the connecting arm; and

two ends of the repair member corresponding to the connecting arm are fixed to the corresponding first connection point and the corresponding repair part, respectively.

8. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 6, wherein the repair member has an elongated strip structure and one repair member is provided; an end of the repair member is capable of extending through all of the repair parts in sequence and being connected to the self-expansion stent.

9. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 6, wherein the repair member has an elongated strip structure; and the number of the repair members is equal to the number of the elastic positioning members; and an end of each of the repair members extends through the repair parts on the same U-shaped portion in sequence and is connected to the self-expansion stent.

10. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 7, further comprising a protective member connected to an outer side of a middle part of the U-shaped portion by suturing, wrapping, or winding.

11. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 10, wherein the repair member and the protective member are polyester cloth or a bovine pericardium.

12. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 2, wherein the outflow end release member comprises a release portion and support arms located on two sides of the release portion; the support arm is connected to the self-expansion stent at the second connection point; two second connection points of the same outflow end release member are located at vertexes of different outflow end grids; and the release portion extends away from the inflow end.

13. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 12, wherein a release grid is enclosed by the support arms of each outflow end release member and adjacent outer sides of two outflow end grids; an end of the valve leaflet adjacent to the inflow end is sutured to the valve skirt in a circumferential direction, and another end of the valve leaflet is connected to the valve skirt at a position inside the release grid.

14. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 13, wherein a hole size of the through-hole gradually becomes smaller and then remains unchanged or gradually becomes larger in a first direction, and the release grid is bent toward an inner side of the through-hole in the first direction.

15. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 13, wherein the elastic positioning member and the self-expansion stent are formed by cutting a nickel-titanium alloy pipe in a single pass, followed by thermoforming.

16. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 1, wherein a sidewall of the self-expansion stent adjacent to the inflow end is formed by a plurality of inflow end grids arranged in an annular shape; and the inflow end release member comprises an elliptical structure connected to a vertex of one of the inflow end grids.

17. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 16, wherein a penetration hole is formed at a middle part of the elliptical structure in a radial direction of the through-hole.

18. The transcatheter aortic valve replacement apparatus having bilateral locking function according to claim 16, wherein an extension line of a short axis of the elliptical structure is parallel to an axis of the through-hole, and an extension line of a long axis of the elliptical structure is located on a plane tangent to the corresponding inflow end grid.
